18.4: A program of differential gene expression leads to the different cell types in a multicellular organism

In the embryonic development of multicellular organisms, a fertilized egg (a zygote) gives rise to cells of many different types, each with a different structure and corresponding function. Typically, cells are organized into tissues, tissues into organs, organs into organ systems, and organ systems into the whole organism. Thus, any developmental program must produce cells of different types that form higher-level structures arranged in a particular way in three dimensions. The processes that occur during animal development are detailed in Chapter 47; in this chapter, we focus on the program of regulation of gene expression that orchestrates development, using a few animal species as examples.

A Genetic Program for Embryonic Development

The photos in **Figure 18.16** illustrate the dramatic difference between a frog zygote (fertilized egg) and the tadpole it becomes. This remarkable transformation results from three interrelated processes: cell division, cell differentiation, and morphogenesis. Through a succession of mitotic cell divisions, the zygote gives rise to a large number of cells. Cell division alone, however, would merely produce a great ball of identical cells, nothing like a tadpole. During embryonic development, cells not only increase in number, but also undergo cell **differentiation**, the process by which cells become specialized in structure and function. Moreover, the different kinds of cells are not randomly distributed but are organized into tissues and organs in a particular three-dimensional arrangement. The physical processes that give an organism its shape constitute **morphogenesis**, the development of the form of an organism and its structures.

Figure 18.16

From fertilized egg to animal: What a difference four days makes.

Figure a is fertilized eggs of a frog on a 1 millimeter scale. Figure b is a newly hatched tadpole on a 2 millimeter scale.

*It takes just four days for cell division, differentiation, and morphogenesis to transform each of the fertilized frog eggs shown in **(a)** into a tadpole like the one in **(b)**.*

All three processes are rooted in cellular behavior. Even morphogenesis, the shaping of the organism, can be traced back to changes in the shape, motility, and other characteristics of the cells that make up various

regions of the embryo. As you have seen, the activities of a cell depend on the genes it expresses and the proteins it produces. Almost all cells in an organism have the same genome; therefore, differential gene expression results from the genes being regulated differently in each cell type.

In Figure 18.12, you saw a simplified view of how differential gene expression occurs in two cell types, a liver cell and a lens cell. Each of these fully differentiated cells has a particular mix of specific transcription factor activators that turn on the collection of genes whose products are required in the cell. The fact that both cells arose through a series of mitoses from a common fertilized egg inevitably leads to a question: How do different sets of activators come to be present in the two cells?

It turns out that materials placed into the egg by maternal cells set up a sequential program of gene regulation that is carried out as embryonic cells divide, and this program coordinates cell differentiation during embryonic development. To understand how this works, we will consider two basic developmental processes. First, we'll explore how cells that arise from early embryonic mitoses develop the differences that start each cell along its own differentiation pathway. Second, we'll see how cellular differentiation leads to one particular cell type, using muscle development as an example.

Cytoplasmic Determinants and Inductive Signals

What generates the first differences among cells in an early embryo? And what controls the differentiation of all the various cell types as development proceeds? You can probably deduce the answer: The specific genes expressed in any particular cell of a developing organism determine its path. Two sources of information, used to varying extents in different species, "tell" a cell which genes to express at any given time during embryonic development.

One important source of information early in development is the egg's cytoplasm, which contains both RNA and proteins encoded by the mother's DNA. The cytoplasm of an unfertilized egg is not homogeneous. Messenger RNAs, proteins, other substances, and organelles are distributed unevenly in the unfertilized egg, and this unevenness has a profound impact on the development of the future embryo in many species. Maternal substances in the egg that influence the course of early development are called **cytoplasmic determinants(Figure 18.17a)**. After fertilization, early mitotic divisions distribute the zygote's cytoplasm into separate cells. The nuclei of these cells may thus be exposed to different cytoplasmic determinants, depending on which portions of the zygotic cytoplasm a cell received. The combination of cytoplasmic determinants in a cell helps determine its developmental fate by regulating expression of the cell's genes during the course of cell differentiation.

Figure 18.17

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Sources of developmental information for the early embryo.
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Sources of developmental information for the early embryos.

Parts are as follows.

Part a. Cytoplasmic determinants in the egg. Unfertilized egg with molecules of two different cytoplasm determinants (nucleus also labeled) shows one type of determinant molecule concentrated in the cytoplasm at the left of the cell and the other in the right. Egg undergoes fertilization by a sperm, producing the zygote (fertilized egg) with determinant molecules still unevenly distributed. Zygote contains both types of cytoplasmic determinants that were present in the egg. Zygote undergoes mitotic cell division into a two-celled embryo; one cell (with nucleus) contains almost entirely one type of cytoplasmic determinant, and the other cell (with nucleus) contains almost entirely the other type of cytoplasmic determinant. Summary: The unfertilized egg has molecules in its cytoplasm, encoded by the mother's genes, that influence development. Many of these cytoplasmic determinants, like the two shown here, are unevenly distributed in the egg. After fertilization and mitotic division, the cell nuclei of the embryo are exposed to different sets of cytoplasmic determinants and, as a result, express different genes.

Part b. Induction by nearby cells. Early embryo (32 cells). Cells at the bottom of the early embryo release molecules that signal (induce) nearby cells to change their gene expression. Diagram zooms in on induction in the cells at the bottom of the embryo: Signaling molecules leave lower cell and bind to signal receptors in the cell above. This begins a signal transduction pathway that sends signals to the cell nucleus of the upper cell.

The other major source of developmental information, which becomes increasingly important as the number of embryonic cells increases, is the environment around a particular cell. Most influential are the signals conveyed to an embryonic cell from other embryonic cells in the vicinity, including contact with cell-surface molecules on neighboring cells and the binding of growth factors secreted by neighboring cells (see Concept 11.1). Such signals cause changes in the target cells, a process called **induction(Figure 18.17b)**. The molecules that transmit these signals within the target cell are cell-surface receptors and other signaling pathway proteins. In general, the signal sends a cell down a specific developmental path by causing changes in its gene expression that lead to observable cellular changes. Thus, interactions between embryonic cells help induce differentiation into the many specialized cell types making up a new organism.

Sequential Regulation of Gene Expression During Cellular Differentiation

The earliest changes that set a cell on its path to specialization are subtle ones, showing up only at the molecular level. Before biologists knew much about the molecular changes occurring in embryos, they coined the term **determination** to refer to the point at which an embryonic cell is irreversibly committed to becoming a particular cell type. Once it has undergone determination, an embryonic cell can be experimentally placed in another location in the embryo and it will still differentiate into the cell type that is its normal fate. Differentiation, then, is the process by which a cell attains its determined fate. As the tissues and organs of an embryo develop and their cells differentiate, the cells become more noticeably different in structure and function.

Today we understand determination in terms of molecular changes that result in observable cell differentiation, marked by the expression of genes for *tissue-specific proteins*. Such proteins are found only in a specific cell type and give the cell its characteristic structure and function. The first sign of differentiation is the appearance of mRNAs for tissue-specific proteins. Later, differentiation is observable with a microscope as changes in cellular structure. On the molecular level, different sets of genes are sequentially expressed in a regulated manner as new cells arise when their precursors divide. Multiple steps in gene expression may be regulated during differentiation, transcription being the most common. In the fully differentiated cell, transcription remains the principal regulatory point for maintaining appropriate gene expression.

Differentiated cells are specialists at making tissue-specific proteins. For example, as a result of transcriptional regulation, liver cells specialize in making albumin, and lens cells specialize in making crystallin (see Figure 18.12). Skeletal muscle cells in vertebrates are another instructive example. Each of these cells is a long fiber containing many nuclei within a single plasma membrane. Skeletal muscle cells have high concentrations of muscle-specific versions of the contractile proteins myosin and actin, as well as membrane receptor proteins that detect signals from nerve cells.

Muscle cells develop from embryonic precursor cells that have the potential to develop into a number of cell types, including cartilage cells and fat cells, but particular conditions commit them to becoming muscle cells. Although the committed cells appear unchanged under the microscope, determination has occurred, and they are now a cell type called *myoblasts*. Eventually, myoblasts start to churn out large amounts of muscle-specific proteins and fuse to form mature, elongated, multinucleate skeletal muscle cells.

Researchers have worked out what happens at the molecular level during muscle cell determination. To do so, they grew embryonic precursor cells in culture and analyzed them using molecular techniques you will learn about in Concepts 20.1 and 20.2. In a series of experiments, they isolated different genes, caused each to be expressed in a separate embryonic precursor cell, and then looked for differentiation into myoblasts and muscle cells. In this way, they identified several so-called “master regulatory genes” whose protein

products commit the cells to becoming skeletal muscle cells. Thus, in the case of muscle cells, the molecular basis of determination is the expression of one or more of these master regulatory genes.

To understand more about how determination occurs in muscle cell differentiation, let's focus on the master regulatory gene called *myoD*. The *myoD* gene deserves its designation as a master regulatory gene. Researchers have shown that the MyoD protein it encodes is capable of changing some kinds of fully differentiated nonmuscle cells, such as fat cells and liver cells, into muscle cells. Why doesn't MyoD work on *all* kinds of cells? One likely explanation is that activation of muscle-specific genes is not solely dependent on MyoD but requires a particular *combination* of regulatory proteins, some of which are lacking in cells that do not respond to MyoD.

What is the molecular basis for muscle cell differentiation? The MyoD protein is a transcription factor (see Figure 18.10) that binds to specific control elements in the enhancers of various target genes and stimulates their expression **(Figure 18.18)**. Some target genes for MyoD encode still other muscle-specific transcription factors. MyoD also stimulates expression of the *myoD* gene itself, an example of positive feedback that perpetuates MyoD's effect in maintaining the cell's differentiated state. Presumably, all the genes activated by MyoD have enhancer control elements recognized by MyoD and are thus coordinately controlled. Finally, the secondary transcription factors activate the genes for proteins such as myosin and actin that confer the unique properties of skeletal muscle cells.

Figure 18.18

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Determination and differentiation of muscle cells.
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Determination and differentiation of muscle cells.

Skeletal muscle cells arise from embryonic cells as a result of changes in gene expression. (In this depiction, the process of gene activation is greatly simplified.)

1. Determination. Diagram zooms in on the nucleus of an embryonic precursor cell. In the D N A is a master regulatory gene myo D that is turned off. Other muscle specific genes on other D N A molecules are also turned off. Arrows from embryonic precursor cell flow to a myoblast (determined), and diagram zooms in on the myoblast nucleus. Signals from other cells lead to the activation of the myo D gene, and the cell makes myo d protein, a specific transcription factor that acts as an activator. The other muscle-specific genes on other D N A molecules are still off. The cell, now called a myoblast, is irreversibly committed to becoming a skeletal muscle cell. 2 Differentiation. Diagram shows myoblast flowing to part of a muscle fiber (fully differentiated cell). myo d protein stimulates the myo d gene further and activates genes encoding other muscle-

specific transcription factors, which in turn activate genes for muscle proteins as follows: Master regulatory gene myo d is transcribed into m R N A and myo d translated. myo d then further stimulates the myo d gene, and it also stimulates transcription and translation on another muscle-specific genes on another D N A molecule. A transcription factor is produced, which in turn stimulates transcription and translation of other muscle-specific genes on other D N A molecules, and myosin, other muscle proteins and cell-cycle blocking proteins are produced. myo d also turns on genes that block the cell cycle, thus stopping cell division. The non-dividing myoblasts fuse to become mature multi-nucleate muscle cells, also called muscle fibers.

WHAT IF? *What would happen if a mutation in the* myoD *gene resulted in the production of an altered MyoD protein that could not activate the* myoD *gene?*

*For suggested answer, see*Appendix A*.*

The determination and differentiation of other kinds of tissues may play out in a similar fashion. Experimental results support the idea that master regulatory proteins like MyoD might function by opening the chromatin in certain regions. This allows access to transcription machinery for activation of the next set of cell-type-specific genes.

Pattern Formation: Setting Up the Body Plan

We have now seen how different programs of gene expression that are activated in the fertilized egg can result in differentiated cells and tissues. But for the tissues to function effectively in the organism as a whole, the organism's *body plan*—its overall three-dimensional arrangement—must be established and superimposed on the differentiation process. Let's look at the molecular basis for the establishment of the body plan, using the well-studied fruit fly *Drosophila melanogaster* as an example.

Cytoplasmic determinants and inductive signals both contribute to spatially organizing the tissues and organs of an organism in their characteristic places. This developmental process is referred to as **pattern formation**.

Just as the locations of the front, back, and sides of a new building are determined before construction begins, pattern formation in animals begins in the early embryo, when the major axes of an animal are established. In a bilaterally symmetrical animal, the relative positions of head and tail, right and left sides, and back and front—the three major body axes—are set up before the organs appear. The molecular cues that control pattern formation, collectively called **positional information**, are provided by cytoplasmic determinants and inductive signals (see Figure 18.17). These cues tell a cell its location relative to the body axes and to neighboring cells, and determine how the cell and its descendants will respond to future molecular signals.

During the early 20th century, classical embryologists made detailed anatomical observations of embryonic development in a number of species and performed experiments in which they manipulated embryonic tissues. Although this research laid the groundwork for understanding the mechanisms of development, it did not reveal the specific molecules that guide development or determine how patterns are established.

In the 1940s, scientists began using the genetic approach—the study of mutants—to investigate *Drosophila* development. That approach has had spectacular success. These studies have established that genes control development and have led to an understanding of the key roles that specific molecules play in defining position and directing differentiation. By combining anatomical, genetic, and biochemical approaches to the study of *Drosophila* development, researchers have discovered developmental principles common to many other species, including humans.

The Life Cycle of *Drosophila*

Fruit flies and other arthropods have a modular construction, an ordered series of segments. These segments make up the body's three major parts: the head, the thorax (the mid-body, from which the wings and legs extend), and the abdomen **(Figure 18.19a)**. Like other bilaterally symmetrical animals, *Drosophila* has an anterior-posterior (head-to-tail) axis, a dorsal-ventral (back-to-belly) axis, and a right-left axis. In *Drosophila*, cytoplasmic determinants that are localized in the unfertilized egg provide positional information for the placement of anterior-posterior and dorsal-ventral axes even before fertilization. We'll focus here on the molecules involved in establishing the anterior-posterior axis as a case in point.

Figure 18.19

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Key events in *Drosophila* development.
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Key events in Drosophila development.

Parts are as follows.

Part a. Adult. The adult fly is segmented, and multiple segments make up each of the three main body parts, head, thorax, and abdomen. Arrows show the body axes. Body plan of the fly, left to right and color coded: head (green) is attached to thorax (brown), followed by abdomen (orange). These colors correspond to elements in the rest of the diagram. (Scale: 0.5 millimeters). Body axes: Anterior (head) and posterior (abdomen) axis (horizontal); Dorsal (back) and ventral (underside) axis (vertical); right-left axis (arrows between dorsal and posterior and anterior and ventral).

Part b. Development from egg to larva.

- 1. Developing Egg. Egg (with nucleus) surrounded by support cells (follicle cells) within one of the mother's ovaries. Nurse cells labeled near the egg cell.
- 2. Mature, unfertilized egg. Developing egg enlarges as nutrients and m R N As are supplied to it by other support cells (nurse cells), which shrink. Egg shell labeled.
- 3. Fertilized egg. The egg is fertilized within the mother and then laid.
- 4. Egg develops into segmented embryo. Green region corresponds to head in adult fly; brown behind green to thorax; followed by orange to abdomen. Segments visible. Scale: 0.1 millimeters.
- 5. Larva: three stages. Third stage forms pupa (not shown) within which the larva metamorphoses into the adult shown in a. Segmented body and colored coded regions as in 4.

The *Drosophila* egg develops in one of the female's ovaries, next to the nurse cells, which supply the egg with nutrients, mRNAs, and other substances needed for development. The egg and nurse cells are surrounded by follicle cells, which make the eggshell (**Figure 18.19b**, top). After fertilization and laying of the egg, embryonic development results in the formation of a segmented larva, which goes through three larval stages. Then, in a process much like that by which a caterpillar becomes a butterfly, the fly larva forms a pupa in which it metamorphoses into the adult fly pictured in Figure 18.19a.

Genetic Analysis of Early Development: *Scientific Inquiry*

Edward B. Lewis was a visionary American biologist who, in the 1940s, first showed the value of the genetic approach to studying embryonic development in *Drosophila*. Lewis studied bizarre mutant flies with developmental defects that led to extra wings or legs in the wrong place **(Figure 18.20)**. He located the mutations on the fly's genetic map, thus connecting the developmental abnormalities to specific genes. This research supplied the first concrete evidence that genes somehow direct the developmental processes studied by embryologists. The genes Lewis discovered, called **homeotic genes**, are regulatory genes that control pattern formation in the fly.

Figure 18.20

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Abnormal pattern formation in *Drosophila*.
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Two flies. Wild type fly: eyes with antennae in between. Mutant fly: eyes with legs in between.

Mutations in homeotic genes cause misplacement of structures in an animal, such as the legs extending from the mutant fly's head in place of antenna (colorized SEMs).

Further insight into pattern formation during early embryonic development did not come for another 30 years, when two researchers in Germany, Christiane Nüsslein-Volhard and Eric Wieschaus, set out to identify *all* the genes that affect segment formation in *Drosophila*. The project was daunting for three reasons. The first was the sheer number of *Drosophila* genes, now known to total about 14,000. The genes affecting segmentation might be just a few needles in a haystack or might be so numerous and varied that the scientists would be unable to make sense of them. Second, mutations affecting a process as fundamental as segmentation would surely be **embryonic lethals**, mutations with phenotypes causing death at the embryonic or larval stage. Because organisms with embryonic lethal mutations never reproduce, they cannot be bred for study. The researchers dealt with this problem by looking for recessive mutations, which can be propagated in heterozygous flies that act as genetic carriers. Third, cytoplasmic determinants in the egg were known to play a role in axis formation, so the researchers knew they would have to study the mother's genes as well as those of the embryo. It is the mother's genes that we will discuss further as we focus on how the anterior-posterior body axis is set up in the developing egg.

Nüsslein-Volhard and Wieschaus began their search for segmentation genes by exposing flies to mutagenic agents and scanning their descendants for dead embryos or larvae with abnormal segmentation or other defects. For example, to find genes that might set up the anterior-posterior axis, they looked for embryos or larvae with abnormal ends, such as two heads or two tails, predicting that such abnormalities would arise from mutations in maternal genes required for correctly setting up the offspring's head or tail end.

Using this approach, Nüsslein-Volhard and Wieschaus eventually identified about 1,200 genes essential for pattern formation during embryonic development. Of these, about 120 were essential for normal segmentation patterns. Next, the researchers were able to group these segmentation genes by general function and to isolate many of them for further study. The result was a detailed molecular understanding of the early steps in pattern formation in *Drosophila*.

When the results of Nüsslein-Volhard and Wieschaus were combined with Lewis's earlier work, a coherent picture of *Drosophila* development emerged. In recognition of their discoveries, the three researchers were awarded a Nobel Prize in 1995. Let's consider a specific example of the genes that Nüsslein-Volhard, Wieschaus, and co-workers found.

Axis Establishment

As we mentioned earlier, cytoplasmic determinants in the egg are the substances that initially establish the axes of the *Drosophila* body. These substances are encoded by genes of the mother, fittingly called maternal

effect genes. A gene classified as a **maternal effect gene** is one that, when mutant in the mother, results in a mutant phenotype in the offspring, regardless of the offspring's own genotype. In fruit fly development, the mRNA or protein products of maternal effect genes are placed in the egg while it is still in the mother's ovary. When the mother has a mutation in such a gene, she makes a defective gene product (or none at all), and her eggs are abnormal; when these eggs are fertilized, they fail to develop properly.

Because maternal effect genes control the orientation (polarity) of the egg and consequently that of the fly, they are also called *egg-polarity genes*. Two groups of these genes set up the anterior-posterior and dorsal-ventral axes of the embryo. Like mutations in segmentation genes, mutations in maternal effect genes are generally embryonic lethals.

Bicoid: A Morphogen That Determines Head Structures

To see how maternal effect genes determine the body axes of the offspring, we will focus on one such gene, called *bicoid*, a term meaning "two-tailed." An embryo or larva whose mother has two mutant *bicoid* alleles lacks the front half of its body and has posterior structures at both ends **(Figure 18.21)**. This phenotype suggested to Nüsslein-Volhard and her colleagues that the product of the mother's *bicoid* gene is essential for setting up the anterior end of the fly and might be concentrated at the future anterior end of the embryo. This hypothesis is an example of the *morphogen gradient hypothesis* first proposed by embryologists a century ago, in which gradients of substances called **morphogens** establish an embryo's axes and other features of its form.

Figure 18.21

Effect of the *bicoid* gene on *Drosophila* development.

Two fly larva.

Wild-type larva: head, tail and segments labeled from head to tail: T1, T2, T3, A1, A2, A3, A4, A5, A6, A7, A8. A6-A8 near the tail are highlighted. Mutant larva (bicoid) with tails at both ends. All labeled segments highlighted. From left to right: A8, A7, A6, A7, A8.

A wild-type fruit fly larva has a head, three thoracic (T) segments, eight abdominal (A) segments, and a tail. A larva whose mother has two mutant alleles of the *bicoid* gene has two tails and lacks all anterior structures (LMs).

DNA technology and other modern biochemical methods enabled the researchers to test whether the *bicoid* product, a protein called Bicoid, is in fact a morphogen that determines the anterior end of the fly. First, they asked whether the location of the mRNA and protein products of this gene in the egg was consistent with the hypothesis. They found that *bicoid* mRNA is highly concentrated at the extreme anterior end of the mature egg **(Figure 18.22)**. After the egg is fertilized, the mRNA is translated into protein. The Bicoid protein then

diffuses from the anterior end toward the posterior, resulting in a gradient of protein within the early embryo, most highly concentrated at the anterior end. These results are consistent with the hypothesis that Bicoid protein specifies the fly's anterior end. To test this more specifically, scientists injected pure *bicoid* mRNA into various regions of early embryos. The protein that resulted from its translation caused anterior structures to form at the injection sites.

Figure 18.22

Inquiry: Could Bicoid be a morphogen that determines the anterior end of a fruit fly?

Experiment

Using a genetic approach to study *Drosophila* development, Christiane Nüsslein-Volhard and colleagues at two research institutions in Germany analyzed expression of the *bicoid* gene. The researchers hypothesized that *bicoid* normally codes for a morphogen that specifies the head (anterior) end of the embryo. To begin to test this hypothesis, they used molecular techniques to determine whether the mRNA and protein encoded by this gene were found in the anterior end of the fertilized egg and early embryo of wild-type flies.

Results

Bicoid mRNA (dark blue in the light micrographs and drawings) was confined to the anterior end of the unfertilized egg. Later in development, Bicoid protein (dark orange) was seen to be concentrated in cells at the anterior end of the embryo.

A diagram shows bicoid m R N A at the anterior end in a mature unfertilized egg. After fertilization and translation of bicoid m R N A, bicoid protein is concentrated in the anterior end of the early embryo.

Conclusion

The location of *bicoid* mRNA and the diffuse gradient of Bicoid protein seen later are consistent with the hypothesis that Bicoid protein is a morphogen specifying formation of head-specific structures.

Further Reading C. Nüsslein-Volhard et al., Determination of anteroposterior polarity in *Drosophila*, *Science*238:1675–1681 (1987); W. Driever and C. Nüsslein-Volhard, A gradient of Bicoid protein in

Drosophila embryos, *Cell*54:83–93 (1988); T. Berleth et al., The role of localization of *bicoid*RNA in organizing the anterior pattern of the *Drosophila* embryo, *EMBO Journal*7:1749–1756 (1988).

WHAT IF? *The researchers needed further evidence, so they injected* bicoid *mRNA into the anterior end of an egg from a female with a mutation disabling the* bicoid *gene. Given that the hypothesis was supported, what must their results have been?*

*For suggested answer, see*Appendix A.*

Animation

Role of *bicoid* Gene in *Drosophila* Development

image

Play

image

Play

The *bicoid* research was groundbreaking for several reasons. First, it led to the identification of a specific protein required for some of the earliest steps in pattern formation. It thus helped us understand how different regions of the egg can give rise to cells that go down different developmental pathways. Second, it increased our understanding of the mother's critical role in the initial phases of embryonic development. Third, the principle that a gradient of morphogens can determine polarity and position has proved to be a key developmental concept for a number of species, just as early embryologists had hypothesized.

Maternal mRNAs are crucial during development of many species. In *Drosophila*, gradients of specific proteins encoded by maternal mRNAs not only determine the posterior and anterior ends but also establish the dorsal-ventral axis. As the fly embryo grows, it reaches a point when the embryonic program of gene expression takes over, and the maternal mRNAs must be destroyed. Later, positional information encoded by the embryo's genes, operating on an ever finer scale, establishes a specific number of correctly oriented segments and triggers the formation of each segment's characteristic structures. Mutant genes affecting this final step can lead to an abnormal adult pattern, as you saw in Figure 18.20.

Interview

Interview with Nancy Hopkins: Studying the genetic basis of development in zebrafish

Nancy Hopkins.

Evolutionary Developmental Biology (“Evo-Devo”)

EVOLUTION The fly with legs emerging from its head in Figure 18.20 is the result of a single mutation in one gene, a homeotic gene. The gene does not encode any antenna protein, however. Instead, it encodes a transcription factor that regulates other genes, and its malfunction leads to misplaced structures, such as legs instead of antennae. The observation that a change in gene regulation during development could lead to such a fantastic change in body form prompted some scientists to consider whether these types of mutations could contribute to evolution by generating novel body shapes. Ultimately, this line of inquiry gave rise to the field of evolutionary developmental biology, so-called “evo-devo,” which will be further discussed in Concept 21.6.

In this section, we have seen how a carefully orchestrated program of sequential gene regulation controls the transformation of a fertilized egg into a multicellular organism. The program is carefully balanced between turning on the genes for differentiation in the right place and turning off other genes. Even when an organism is fully developed, gene expression is regulated in a similarly fine-tuned manner. In the final section of the chapter, we’ll consider how fine this tuning is by looking at how specific changes in expression of just a few genes can lead to the development of cancer.